Recoverable Hindsight Regret: Calibrated Selective Repair of Trading Policies

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Abstract

A historical price path can reveal that another trading policy would have performed better, without establishing whether that choice was predictable at the time. We examine this distinction under price-taking replay. Frozen policies are compared from a common account state over 60 trading days; past-only utility predictions and an empirical calibrated lower score determine when to take a candidate's current action. Subsequent decisions are reconsidered. Among 24 registered policy/model/split groups, 17 have positive individual 95% paired block-bootstrap intervals for selected-decision continuation advantage. The comparisons are dependent and the intervals are approximate. Executed-path evidence is weaker: one fold-local interval is positive, one is negative, and no continuous selective-repair interval is entirely positive. In controlled synthetic settings, useful decision-time signals yield recoverable opportunities, while clairvoyance-only opportunities lead to abstention. Conditional mathematical results explain the required validity of lower scores and the non-additivity of overlapping continuation advantages. These results support policy-dependent local recoverability under stated replay assumptions; sustained improvement in trading wealth is not established.

1 Introduction

Hindsight can identify decisions for which another action would have performed better, but it does not establish that the mistake was avoidable when the decision was made. A hindsight comparator may exploit information unavailable to any deployable rule. We study the narrower question: which realized policy mistakes remain recoverable from decision-time information?

Trading provides a useful replayable setting. Under a price-taking assumption and a known accounting/execution map, a frozen policy family can be run ex post on the same realized market path. This differs from partial-feedback off-policy evaluation, where rewards for unchosen actions must be inferred (Dudík et al., 2011). Replayability solves the missing-counterfactual problem only under these assumptions; it does not reveal ex ante which continuation will be preferable.

For incumbent policy k and candidate j, let $A_{t,H}^{k\to j}(S_t)$ be their realized post-cost continuation-utility difference over horizon H from the same account state. The information-constrained target

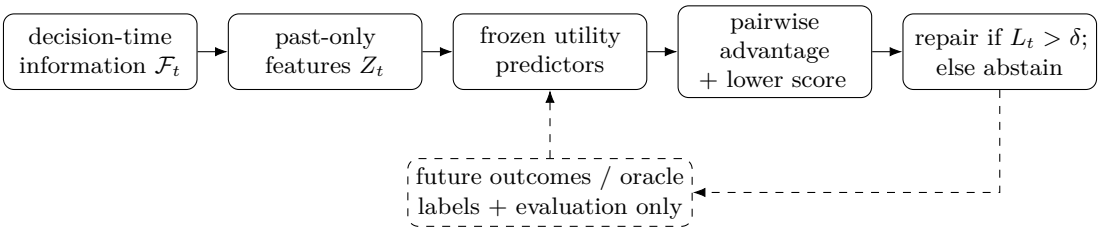


Figure 1: Information boundary of selective repair. Future continuation outcomes may create historical labels and evaluate frozen decisions, but they are not deployment inputs.

is

$$\mu_{t,H}^{k\to j} = \mathbb{E}\Big[A_{t,H}^{k\to j}(S_t) \mid \mathcal{F}_t\Big]. \tag{1}$$

The empirical procedure estimates policy-specific continuation utilities from past-only features, differences the frozen predictions, and intervenes only when a calibration-derived lower score clears a fixed margin. Otherwise it retains the incumbent. Importantly, a 60-day policy-continuation comparison screens a *current* action, while the implemented repaired policy reconsiders its action at subsequent dates. The label is a decision-time ranking surrogate, not an estimate of the isolated causal effect of one switch followed by later reoptimization. Future continuation labels and hindsight-oracle outputs never enter the deployment interface.

The formal results separate hindsight opportunity, conditional advantage, and realized-path performance. Theorem 5.1 characterizes the clairvoyance-only case; Propositions 5.2 and 5.3 give consequences of valid lower scores; and Lemma 5.4 explains why overlapping local continuation gains cannot be added into a wealth claim. The empirical residual-quantile score is not assumed to satisfy the population premises of those propositions. Supporting oracle-gap identities appear in Appendix A.

Across annual walk-forward tests, 17 of 24 registered policy/model/split groups have positive individual intervals for selected-decision continuation advantage. One fold-local executed-path interval is positive, one negative, and no continuous selective-repair interval is entirely positive. All intervals use the specified block-bootstrap approximation, without multiplicity adjustment. Synthetic controls distinguish information-driven recoverability from positive hindsight opportunities without a usable signal.

Contributions. The information-constrained target makes the hindsight oracle a diagnostic comparator rather than an available trading rule. Common-state replay provides cost-inclusive labels for frozen, past-only continuation-utility predictors; a residual-calibrated gate abstains when predicted pairwise advantage is insufficient. The analysis then compares selected-decision continuation outcomes with actual repaired wealth paths, retaining negative cases and synthetic information controls. The empirical procedure is a selective decision screen. Its conditional theory does not certify the implemented multi-candidate selector, and the market results do not demonstrate sustained portfolio improvement.

2 Related Work

Hindsight comparators are standard in online learning and portfolio selection. Universal portfolios compare sequential performance with a fixed portfolio chosen in hindsight (Cover, 1991), while Besbes et al. (2021) study learning when the feedback is the action an informed oracle would have

taken. Here the oracle has a different role: it constructs retrospective continuation comparisons, while the deployable object is a candidate-versus-incumbent continuation advantage measurable from decision-time information.

The setting also differs from standard off-policy evaluation. Contextual-bandit methods such as doubly robust evaluation address missing rewards for actions not taken (Dudík et al., 2011). Under the price-taking and deterministic-accounting assumptions used here, the recorded price path instead permits ex-post replay of every policy in the frozen family from a common account state. The remaining obstacle is informational rather than counterfactual: replay reveals realized advantage after the fact, not whether that advantage was identifiable at time t.

The intervention rule is closest in architecture to safe policy improvement and selective prediction. High-confidence and robust-baseline methods seek improvements while limiting degradation relative to a baseline (Thomas et al., 2015; Ghavamzadeh et al., 2016); SPIBB restricts departures in weakly supported regions (Laroche et al., 2019); and decision-point methods localize improvement to sufficiently supported states (Sharma et al., 2025). Selective classification makes abstention and the risk–coverage trade-off part of the decision rule (Geifman and El-Yaniv, 2017). The distinction here is the screened object: a common-state, pairwise continuation advantage within a frozen policy family, calibrated from historical prediction residuals rather than from visitation support alone. Conformal work motivates careful separation between empirical calibration and formal coverage guarantees (Romano et al., 2019; Gibbs and Candès, 2021); the score in eq. (12) is therefore called a *calibrated lower score*, not an exact conformal or conditional-mean confidence bound.

Finally, inference for repeated financial backtests must respect dependence and specification search. Block resampling provides classical tools for dependent observations (Künsch, 1989; Politis and Romano, 1994); the Reality Check, superior-predictive-ability test, Deflated Sharpe Ratio, and probability of backtest overfitting address related selection concerns (White, 2000; Hansen, 2005; Bailey and López de Prado, 2014; Bailey et al., 2017). The present design freezes the policy/model registry, asset split, execution rule, thresholds, and estimands before the reported inference; it also retains held-out assets and negative controls. These choices do not make the public-market study confirmatory, but they keep the local recoverability claim auditable and distinct from the substantially weaker path-level evidence.

3 Problem Setting and Counterfactual Replay

We consider a price-taking investor and a finite, prespecified family of non-anticipating trading policies evaluated on one realized market path. Future outcomes may be used retrospectively for replay, labels, and evaluation, but every deployed action must remain measurable with respect to the information available when the decision is made.

3.1 Information, state, and execution

Let $X_{1:T} = (X_1, \dots, X_T)$ denote the realized exogenous market path and let $S_t \in \mathcal{S}$ be the account state after the close at time t, including holdings, cash, wealth, and any policy state. Write I_t for the complete decision-time information and $\mathcal{F}_t = \sigma(I_t)$. A deployable policy π is non-anticipating:

$$a_t^\pi = \pi_t(I_t) \in \mathcal{A}, \qquad a_t^\pi \text{ is } \mathcal{F}_t\text{-measurable.} \tag{2}$$

The frozen family is $\Pi_0 = \{\pi^1, \dots, \pi^K\} \subset \Pi_{\mathcal{F}}$.

Decisions are formed after the close at t and executed at the next tradable open. Execution, costs, corporate actions, and marking are represented by a common transition map

$$S_{t+1}^\pi = \Phi_t(S_t^\pi, a_t^\pi, X_{t+1}), \qquad W_t^\pi = w(S_t^\pi). \tag{3}$$

The price-taking assumption is pathwise: the policies do not alter $X_{1:T}$. It does not imply frictionless trading. Transaction costs, cash accrual, holdings, and all later state variables are recomputed under each counterfactual path. The replay argument therefore does not automatically transfer to settings with material market impact, unknown execution response, or strategic counterparties.

Assumption 3.1 (Replay conditions)**.** The realized market path is exogenous to the investor's actions; Φ_t is known; the state transition is deterministic conditional on (S_t, a_t, X_{t+1}); and each policy in Π_0 is deterministic and non-anticipating once its observed history and state are fixed.

Under Assumption 3.1, $(S_0, X_{1:T}, \pi)$ uniquely determines the state, action, and wealth path of any $\pi \in \Pi_0$ by forward recursion. This replay property is a structural feature of the setting, not a claim that off-policy evaluation is unnecessary in markets generally.

3.2 Common-state continuation advantage

Fix a decision time t, horizon H, and common starting state s. Let $W_{t+h}^{k;t,s}$ be wealth obtained by replaying policy π^k from s over the realized future segment. Define the post-cost continuation utility

$$U_{t,H}^{(k)}(s) = \log\left(\frac{W_{t+H}^{k;t,s}}{W_t(s)}\right), \tag{4}$$

and the realized pairwise advantage

$$A_{t,H}^{k\to j}(s) = U_{t,H}^{(j)}(s) - U_{t,H}^{(k)}(s). \tag{5}$$

The common starting state is essential: values beginning from different holdings or wealth states are not comparable measures of the value of changing the continuation policy at time t. The counterfactual in eq. (5) commits to the respective policy for H trading days from that shared state. It is *not* the one-step treatment effect of taking policy j's next action and subsequently reoptimizing; the latter generally has a different continuation distribution and wealth path.

For the empirical hindsight benchmark, the oracle may inspect $X_{t+1:t+H}$ but must still choose within the same frozen continuation family and use the same accounting map:

$$j_{t,H}^{\mathrm{O}}(s) \in \arg\max_{j\in\{1,\dots,K\}} U_{t,H}^{(j)}(s), \qquad U_{t,H}^{(\mathrm{O})}(s) = \max_j U_{t,H}^{(j)}(s). \tag{6}$$

The local hindsight regret of incumbent k is

$$G_{t,H}^{(k)}(s) = U_{t,H}^{(\mathrm{O})}(s) - U_{t,H}^{(k)}(s). \tag{7}$$

The oracle is diagnostic: policy repair uses the pairwise quantity in eq. (5), and candidates remain deployable members of Π_0.

3.3 Deployment boundary

Let $\mathcal{G}_T$ be the ex-post sigma-field generated by the full recorded path and registered covariates, with $\mathcal{F}_t \subseteq \mathcal{G}_T$. The quantities in eqs. (5) and (7) are generally known only after $(t, t+H]$ has elapsed. They may create historical labels and evaluate frozen decisions, but they are not admissible deployment inputs. The theoretical certification developed below is therefore decision-local at the encountered common state; it is not a guarantee for the subsequently altered wealth path, which must be evaluated separately by end-to-end replay.

4 Recoverable Hindsight Regret and Selective Repair

Replay identifies realized continuation advantages only ex post. At decision time the ideal target is the conditional component

$$\mu_{t,H}^{k\to j} = \mathbb{E}\Big[A_{t,H}^{k\to j}(S_t) \mid \mathcal{F}_t\Big]. \tag{8}$$

The implemented predictor uses a registered past-only feature map $Z_t = \psi(I_t)$ with $\mathcal{H}_t = \sigma(Z_t) \subseteq \mathcal{F}_t$. For each policy,

$$m^{(k)}(Z_t) = \mathbb{E}\Big[U_{t,H}^{(k)}(S_t) \mid \mathcal{H}_t\Big], \qquad m^{k\to j}(Z_t) = m^{(j)}(Z_t) - m^{(k)}(Z_t). \tag{9}$$

Thus $\mu_{t,H}^{k\to j}$ denotes the full $\mathcal{F}_t$-conditional target, whereas $m^{k\to j}(Z_t)$ is the target available through the registered feature sigma-field $\mathcal{H}_t$.

4.1 Frozen continuation-utility prediction

Within each training fold, a separate continuation-utility model $\widehat{m}^{(k)}$ is fitted for each policy using replay-generated labels whose full future windows are already available. After training, preprocessing and model parameters are frozen. At test time,

$$\widehat{U}_t^{(k)} = \widehat{m}^{(k)}(Z_t), \qquad \widehat{A}_t^{k\to j} = \widehat{U}_t^{(j)} - \widehat{U}_t^{(k)}. \tag{10}$$

The labels are cost-inclusive because they are generated by the shared execution engine; the repair rule does not subtract a second estimated cost term. Test-time inputs contain only Z_t, current state, and frozen fitted objects—never future continuation labels or oracle outputs. The full-horizon continuation target is used to *rank candidate actions*; the implemented gate does not commit to H days of the selected policy. After each action, both the account state and the decision are updated, so local forecast quality alone cannot establish realized pathwise improvement (Lemma 5.4).

4.2 Calibration and abstention

For ordered pair (k, j), let $\mathcal{C}$ be the calibration observations available before the test period. Define residuals and their empirical lower-tail quantile by

$$r_i^{k\to j} = A_{i,H}^{k\to j}(S_i) - \widehat{A}_i^{k\to j}, \qquad q_\alpha^{k\to j} = Q_\alpha\Big(\{r_i^{k\to j} : i \in \mathcal{C}\}\Big), \tag{11}$$

and the deployed lower score

$$L_t^{k\to j} = \widehat{A}_t^{k\to j} + q_\alpha^{k\to j}. \tag{12}$$

Models and residual quantiles are not updated from test outcomes.

Equation (12) is an empirical lower *prediction* score for realized continuation advantage. It is not a finite-sample lower confidence bound for the conditional mean $\mu_{t,H}^{k\to j}$, and the conditional-mean guarantee in Proposition 5.2 therefore does not automatically apply to this empirical score. The frozen implementation files retain the legacy shorthand *LCB*; the manuscript uses *calibrated lower score*.

Fix incumbent k and let

$$j_t^* \in \arg\max_{j\neq k} L_t^{k\to j}. \tag{13}$$

For deployment margin $\delta \geq 0$, the repaired action is

$$a_t^{\mathrm{rep}(k)} = \begin{cases} a_t^{\pi^{j_t^*}}, & L_t^{k\to j_t^*} > \delta, \\ a_t^{\pi^k}, & \text{otherwise.} \end{cases} \tag{14}$$

The second branch is abstention. The incumbent remains the reference policy at later decisions; any executed intervention changes the actual account state, which is then carried forward by the same execution engine before the next decision is recomputed. A more aggressive stateful meta-selector is retained only as a stress test and is defined in Appendix D.

Selection across candidates. The maximization in eq. (13) chooses among several *pair-specific* lower scores. Fixed-pair calibration does not automatically certify the subsequently selected maximum. If, under the same target law, each of M prespecified candidates separately obeyed $\mathbb{P}(\Delta_j < L_j) \le \alpha_j$, a union bound would control the event that *any* candidate violates its score only by $\sum_{j=1}^{M} \alpha_j$ (capped at one), not automatically by a single α_j. A stronger simultaneous or selection-conditional guarantee would require additional assumptions and calibration. No such guarantee is asserted for the implemented historical residual-quantile scores.

5 Information-Constrained Guarantees

The main theory separates two kinds of lower score that should not be conflated. A *conditional-mean certificate* satisfies a pointwise relation to $\mu_t = \mathbb{E}[\Delta_t \mid \mathcal{F}_t]$; a *predictive lower bound* controls the realized event $\{\Delta_t < L_t\}$. The empirical residual-quantile score in eq. (12) is motivated by the second object and is not asserted to satisfy the first. The propositions below concern a *fixed* candidate comparison or require their stated validity event to hold *simultaneously*; they are not unconditional certificates for the empirical multi-candidate selection rule. Standard oracle-gap and information-refinement identities are recorded in Appendix A.

5.1 Clairvoyance-only opportunity

Theorem 5.1 (Clairvoyance-only opportunity is not deployably recoverable)**.** *Let Δ_t be integrable, $\delta \in \mathbb{R}$, and let $d_t \in [0,1]$ be any $\mathcal{F}_t$-measurable deployment rule. If*

$$\mathbb{E}[\Delta_t \mid \mathcal{F}_t] \le \delta \quad a.s., \tag{15}$$

then

$$\mathbb{E}[d_t(\Delta_t - \delta)] \le 0. \tag{16}$$

If, in particular, Δ_t is independent of $\mathcal{F}_t$, $\mathbb{E}[\Delta_t] = \delta$, and $\mathbb{P}(\Delta_t > \delta) > 0$, every deployable gate has zero expected net advantage while the clairvoyant rule $d_t^{\mathrm{O}} = \mathbf{1}\{\Delta_t > \delta\}$ has strictly positive expected net advantage.

Proof. Because d_t is $\mathcal{F}_t$-measurable, $\mathbb{E}[d_t(\Delta_t - \delta)] = \mathbb{E}[d_t\{\mathbb{E}[\Delta_t \mid \mathcal{F}_t] - \delta\}] \le 0$. The independence case gives equality for deployable gates, while the clairvoyant gain is $\mathbb{E}[(\Delta_t - \delta)_+] > 0$. □

This is the identification target of the clairvoyance-only synthetic world: positive hindsight opportunity need not imply any legal ex-ante deployment opportunity.

5.2 Conditional selective repair

For a fixed pair, write $\mu_t = \mathbb{E}[\Delta_t \mid \mathcal{F}_t]$. The Bayes binary gate for the net objective $\mathbb{E}[d_t(\Delta_t - \delta)]$ is $d_t^* = \mathbf{1}\{\mu_t > \delta\}$, up to ties.

Proposition 5.2 (Conditional-mean lower-score certification)**.** *Let $\mathcal{T}$ be a finite set of decision times. Suppose L_t and $b_t \geq 0$ are $\mathcal{F}_t$-measurable and, on an event $\mathcal{E}$,*

$$L_t \leq \mu_t \leq L_t + b_t \quad \text{for every } t \in \mathcal{T}. \tag{17}$$

Define $\widehat{d}_t = \mathbf{1}\{L_t > \delta\}$. On $\mathcal{E}$, (i) $\widehat{d}_t \leq d_t^*$, so there are no false-positive interventions relative to the Bayes conditional-mean gate; and (ii) every missed Bayes intervention satisfies $\delta < \mu_t \leq \delta + b_t$, with

$$\sum_{t\in\mathcal{T}} (d_t^* - \widehat{d}_t)(\mu_t - \delta) \leq \sum_{t\in\mathcal{T}} b_t \mathbf{1}\{\delta < \mu_t \leq \delta + b_t\}. \tag{18}$$

Proof. If $L_t > \delta$, then (17) gives $\mu_t > \delta$. If $d_t^* = 1$ but $\widehat{d}_t = 0$, then $\mu_t > \delta$, $L_t \leq \delta$, and $\mu_t \leq L_t + b_t \leq \delta + b_t$. Summing the resulting pointwise bound gives (18). □

This proposition is conditional on an actual lower certificate for the conditional mean; no such finite-sample certificate is claimed for the empirical residual score in eq. (12).

5.3 Predictive lower bounds and harm–coverage

Proposition 5.3 (Harm–coverage relation)**.** *Let Δ be realized candidate-versus-incumbent advantage on a future decision and suppose a lower prediction score L satisfies, under the target test law,*

$$\mathbb{P}(\Delta < L) \leq \alpha. \tag{19}$$

For deployment event $D = \{L > \delta\}$ with coverage $C = \mathbb{P}(D) > 0$,

$$\mathbb{P}(\Delta < \delta \mid D) \leq \min\left\{1, \frac{\alpha}{C}\right\}. \tag{20}$$

At $\delta = 0$, this is a bound on realized harm among deployed decisions.

Proof. On D, the event $\{\Delta < \delta\}$ implies $\Delta < L$; hence $\mathbb{P}(\Delta < \delta, D) \leq \alpha$. Divide by C and truncate at one. □

The premise in (19) is a population validity condition for the specified score and candidate, not a consequence of observing nominal calibration coverage. Dependence and calibration-to-test shift can invalidate it; choosing the maximum of several pairwise scores does not preserve the same α without additional simultaneous control. Appendix A gives a simple total-variation sensitivity extension.

5.4 Local-to-path non-additivity

Lemma 5.4 (Overlapping continuation windows)**.** *In the additive representation $\Delta_{t,H} = \sum_{s=t+1}^{t+H} g_s$, for deployment times $\mathcal{D}$,*

$$\sum_{t\in\mathcal{D}} \Delta_{t,H} = \sum_s m_s(\mathcal{D}) g_s, \qquad m_s(\mathcal{D}) = \#\{t \in \mathcal{D} : t < s \leq t + H\}. \tag{21}$$

Thus the same one-step increment can enter many local continuation labels.

Proof. Interchange the two sums and collect the coefficient multiplying each g_s. □

The actual trading system is more path dependent: each intervention changes later holdings, wealth, and turnover costs. Consequently, positive average local continuation advantage cannot be summed into terminal-wealth improvement. End-to-end path utility must be measured by replaying the repaired policy itself.

Table 1: Frozen real-data evaluation protocol.

Component	Frozen specification
Universe / split	Six U.S. equities; development: AAPL, AMZN, MSFT, XOM; held out from all fitting: JPM, WMT.
Policies	Buy-and-hold, SMA(20,100), momentum(126), Donchian(55,20), RSI(14,30,55), volatility target(15%,20).
Clock / accounting	Signal at close t, execution at next tradable open; common shares-and-cash engine; 5 bps per side.
Continuation labels	$H = 60$ trading days; common starting-exposure grid $\{0, .25, .5, .75, 1\}$; post-cost log-wealth utility.
Models / gate	Policy-specific Ridge and histogram gradient boosting; calibration residual lower quantile $\alpha = .10$; deployment margin $\delta = 0$.
Walk-forward	Annual tests 2016–2025; expanding training, one-year calibration, one-year test; 60-day label purge plus five-day embargo.
Inference	Paired circular moving-block bootstrap; 20-day blocks, 1000 resamples, 95% percentile intervals.

6 Empirical Design and Evaluation Protocol

The real-data design is frozen before the results in Section 7. Table 1 gives the registered specification; exact hyperparameters, feature definitions, seed, and file registry are in Appendix C.

6.1 Data, chronology, and leakage control

Daily public-market observations begin in 2005. The supplied Phase-13 download code requests daily historical bars and corporate-action fields using the Python `yfinance` interface (Yahoo Finance data access); the code records request options and download-time metadata per symbol. The price-taking replay uses unadjusted execution prices, corporate actions, cash accrual, and a common proportional transaction-cost rule. This is a public, potentially vendor-revised data source rather than an institutional point-in-time stock universe; the market experiment is methodological, not a market-wide profitability or investability study. The present manuscript does not claim to have independently revalidated the archived prices against an institutional point-in-time feed. The development assets are AAPL, AMZN, MSFT, and XOM; JPM and WMT never enter training or calibration.

Ten annual folds cover 2016–2025. The first trains through 2014, calibrates on 2015, and tests on 2016; training expands thereafter. A continuation label becomes eligible only after its full 60-day future window is observed. Any observation whose label crosses the next information boundary is purged, followed by a five-trading-day embargo. Feature preprocessing, policy-specific utility models, and ordered-pair residual quantiles are fitted only on eligible pre-test data and then frozen. Test outcomes do not update fitted objects.

6.2 Frozen deployment and estimands

The registered feature map contains only information observable through date t, including momentum, volatility, drawdown, moving-average, RSI, and current exposure. Ridge and histogram gradient boosting are carried through without model selection on test performance. The deployment interface contains realized history, current state, the frozen transform and predictors, and the calibration-derived residual quantile.

Three estimands remain separate. The *decision-level continuation advantage* is the realized H-day candidate-minus-incumbent utility among interventions, evaluated by policy continuation from a common state; it is a surrogate evaluation of selected decisions, not the incremental wealth effect of the executed one-step intervention. The *fold-local path utility* is the post-cost paired difference in annual log wealth between repaired and baseline paths. The *continuous pseudo-live utility* is the same pathwise quantity on the unreset 2016–2025 sequence, with fitted objects changing only at registered fold boundaries. The meta-selector follows the same chronology but is secondary.

Uncertainty uses paired circular moving-block bootstrap intervals with 20-day blocks and 1000 resamples. Identical resampling indices are applied to synchronized baseline/adaptive quantities. Because 60-day continuation labels overlap, a 20-day block is shorter than the label horizon and need not capture all induced serial dependence; the resulting percentile intervals are approximate and may be too narrow under longer-range dependence. No block-length sensitivity analysis is claimed here. The 17/24 count describes the signs of *separate*, related per-specification intervals under this procedure. It is not a multiplicity-adjusted set of discoveries, a familywise-error-controlled claim, or evidence of a consistently profitable deployable portfolio.

7 Empirical Evidence

Table 2 separates selected-decision continuation outcomes from executed-path performance before examining individual strategies.

Table 2: Evidence hierarchy across the frozen registered specifications. "Positive" and "negative" denote 95% paired block-bootstrap intervals entirely above or below zero.

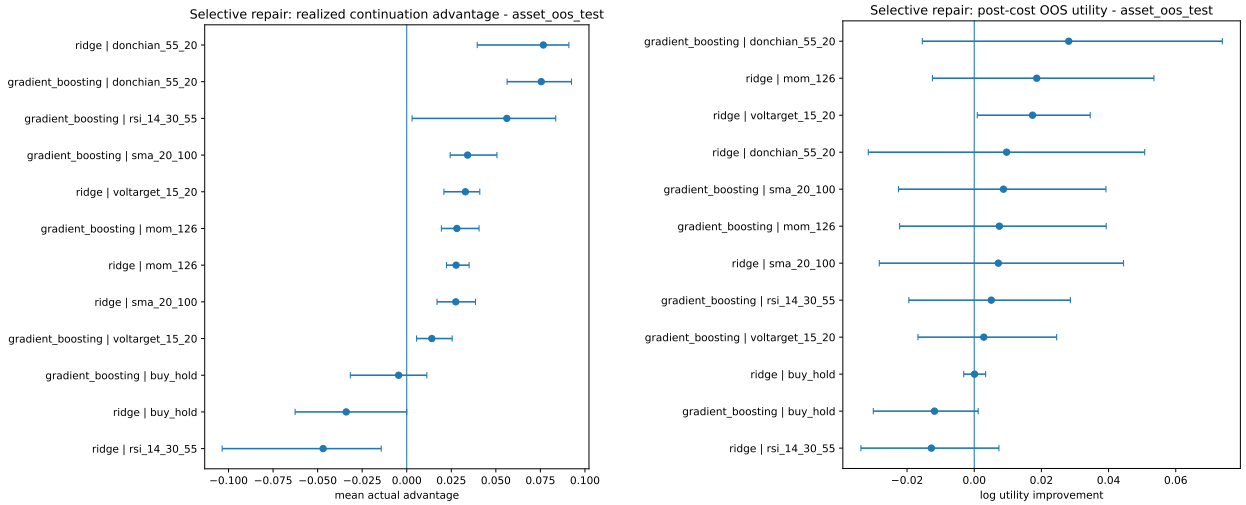
Estimand / strategy	Registered groups	Positive	Negative
Selective repair: decision-level advantage	24	17	2
Selective repair: fold-local path $\Delta \log U$	24	1	1
Selective repair: continuous $\Delta \log U$	24	0	1
Meta-selector: fold-local path $\Delta \log U$	4	0	0
Meta-selector: continuous $\Delta \log U$	4	0	1

7.1 Weak overall prediction and selected-decision outcomes

Absolute continuation-utility and regret regressions frequently have negative test R^2, including weak buy-and-hold branches. No model family was escalated or selected from test performance. What survives out of sample is narrower: some policy-dependent ranking and tail enrichment remain informative enough for the frozen gate to act selectively.

Among the 24 registered policy/model/split groups, 17 have individual 95% paired block-bootstrap intervals for realized decision-level continuation advantage entirely above zero: 9 of 12 held-out-asset groups and 8 of 12 same-asset groups. These comparisons share market histories and overlapping labels; the counts are descriptive rather than simultaneous inferences, and the intervals use the stated 20-day block approximation.

Held-out Donchian repair provides the largest local effects: gradient boosting has mean advantage 0.0755 with 95% interval $[0.0563, 0.0925]$, 10.2% coverage, and 15.2% realized harm; Ridge gives 0.0767 with interval $[0.0395, 0.0910]$. Ridge momentum is more frequently deployed, with mean 0.0277, interval $[0.0224, 0.0350]$, 18.7% coverage, and 8.6% harm. These are selected-decision effects, not estimates of the full oracle gap. The complete held-out table is in Appendix B.



(a) Decision-level realized continuation advantage.

(b) Fold-local post-cost path utility.

Figure 2: Held-out-asset evidence at two estimand levels. Most registered repair settings show positive local continuation evidence, whereas path-level intervals are substantially wider. The panels use the same frozen model/policy registry but represent different estimands.

7.2 Failures and path effects delimit the claim

The gate does not mechanically create positive conditional averages. Held-out Ridge–RSI has mean local advantage -0.0469 with interval $[-0.1036, -0.0143]$ and 80.2% harm among its rare interventions. The buy-and-hold branches provide a second negative control: the same-asset gradient-boosting branch has a negative local interval and is also the only selective-repair branch with a significantly negative fold-local path effect.

Path evidence is much weaker overall. Only 1 of 24 fold-local intervals is entirely above zero and 1 is entirely below zero. The positive case is held-out Ridge volatility targeting, $\Delta \log U = 0.0174$ with 95% interval $[0.0009, 0.0345]$; the negative case is same-asset gradient-boosting buy-and-hold, $\Delta \log U = -0.0189$ with interval $[-0.0338, -0.0046]$. All other fold-local path intervals cross zero. This separation is consistent with Lemma 5.4: overlapping local labels and state-dependent execution prevent decision-level gains from being added into terminal wealth.

7.3 Continuous execution and global selection remain unresolved

None of the 24 continuous selective-repair intervals is entirely above zero. Some point estimates are positive—held-out gradient-boosting Donchian has observed continuous $\Delta \log U = 0.3497$ and Ridge momentum has 0.1860—but the intervals remain wide; held-out Ridge volatility targeting is 0.1654 with interval $[-0.0035, 0.3551]$. Annual effects also reverse across calendar years. The annual heatmap and full continuous forests are therefore reported in Appendix B, rather than presented as a robustness success.

The stateful meta-selector is an intentionally stronger stress test because it requires persistent ranking across the entire policy family. No fold-local meta-selector group has a positive 95% interval, and the continuous same-asset gradient-boosting specification is significantly negative, $\Delta \log U = -0.3016$ with interval $[-0.5459, -0.0210]$. Full details appear in Appendix D.

Section 8 examines the same distinction in controlled settings where the information available at decision time can be varied directly.

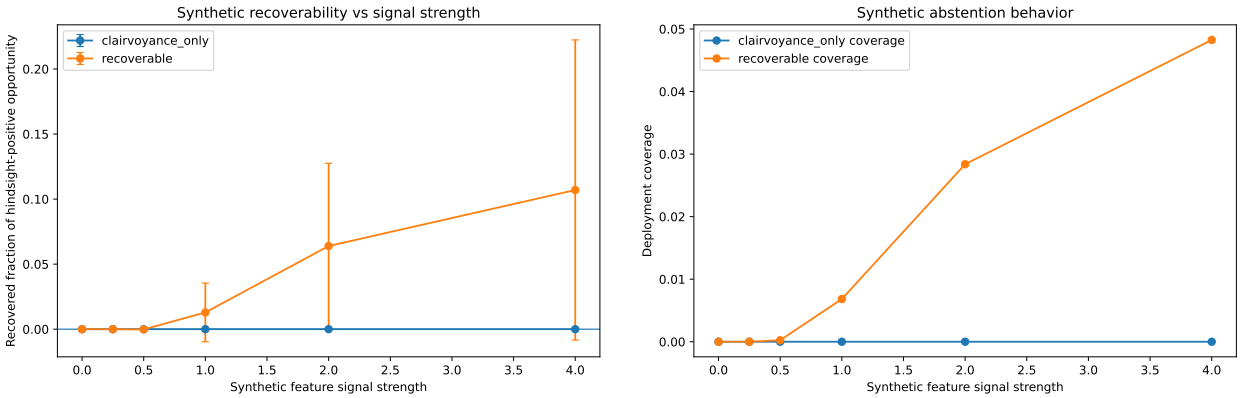
8 Synthetic Identifiability and Falsification

The market data cannot reveal how much hindsight opportunity is recoverable in principle. We therefore use two controlled worlds whose full data-generating processes are given in Appendix B.4. In the *recoverable* world, a decision-time proxy becomes progressively more informative about a persistent regime that affects future returns. In the *clairvoyance-only* world, the same proxy becomes more precise but future innovations—and hence future candidate advantage—remain independent of decision-time information. The first checks that useful information raises recoverability; the second directly targets theorem 5.1.

The synthetic policy pair, execution costs, chronological train/calibration/test split, purge, Ridge fit, and calibration rule are frozen separately from the market experiment. For deployment $d_t(s) = \mathbf{1}\{L_t(s) > 0\}$, we report deployed gain per decision and recovered fraction,

$$G_{\mathrm{dep}}(s) = \mathbb{E}[d_t(s)\Delta_t], \qquad \rho_{\mathrm{syn}}(s) = \frac{G_{\mathrm{dep}}(s)}{\mathbb{E}[(\Delta_t)_+]}, \tag{22}$$

when the denominator is positive, together with rank correlation and deployment coverage. This recovered fraction is an empirical synthetic statistic, not the population oracle-gap ratio in Appendix A.



(a) Recovered fraction of hindsight-positive opportunity. Error bars are across-replicate standard deviations.

(b) Deployment coverage under the same controlled information experiment.

Figure 3: Synthetic information-refinement and falsification checks. In the recoverable world, increasing signal strength reveals more of the regime structure relevant to future policy advantage. In the clairvoyance-only world, hindsight-positive opportunities remain present but the deployment rule abstains throughout the signal grid.

In the recoverable world, mean test Spearman correlation rises from 0.067 at signal $s = 0$ to 0.609 at $s = 4$. Coverage is effectively zero at weak signals, then rises from 0.7% at $s = 1$ to 2.8% at $s = 2$ and 4.8% at $s = 4$. Over the same range, deployed gain per decision reaches 0.001928 and the recovered fraction reaches 0.1070. Finite-sample realized curves need not be monotone at every grid point; the relevant pattern is that nontrivial deployment emerges only when the decision-time proxy carries information about future policy advantage.

The clairvoyance-only world provides the sharper falsification. Average hindsight-positive opportunity remains 0.024478 per decision, yet rank correlations remain near zero, maximum deployment coverage is 0.0000, and deployed gain is zero across the registered signal grid. More

precise but irrelevant information does not manufacture a deployable opportunity. In this world, abstention is the intended outcome; stable positive deployment would instead suggest leakage or overfitting. Full numerical results and the synthetic transition/return specifications are reported in Appendix B.4.

These experiments are an identification check, not a financial return model and not additional evidence for market profitability.

9 Discussion and Limitations

Several policy/model combinations have positive selected-decision continuation contrasts despite weak overall utility prediction. The gate acts on a relatively small subset of states, so broad predictive accuracy and conditional performance among selected decisions need not coincide. The held-out Ridge–RSI and same-asset boosting buy-and-hold results also show that selection can be harmful. A 60-day policy-continuation label compares two hypothetical paths from one state; actual interventions change one action at a time and are reconsidered on subsequent dates. These outcomes address different estimands, as formalized by Lemma 5.4.

The 17/24 positive-interval count is descriptive. The policy specifications share market histories, their continuation labels overlap, and their individual intervals are not adjusted for multiplicity. Paired circular moving-block resampling uses 20-day blocks, shorter than the 60-day continuation horizon. Longer-run dependence could make the reported intervals too narrow; no alternative block-length analysis is reported. Proposition 5.3 presumes predictive validity under the target law, which the historical residual quantiles are not shown to possess. Selection of the maximum among candidate-specific scores additionally lacks simultaneous finite-sample calibration. Abstention therefore cannot be interpreted as a proven safety guarantee.

The market study uses six long-surviving liquid U.S. equities and a fixed six-policy family. Only JPM and WMT are fully held out from fitting, and two held-out asset paths provide limited effective evidence for continuous wealth outcomes. Public vendor data are not an institutional point-in-time universe; asset selection and later data revisions restrict generalization. Replay assumes exogenous prices and a specified accounting engine with proportional trading costs. Market impact, capacity, queue position, and intraday execution uncertainty are not modeled. The experiment characterizes local, policy-dependent decision opportunities in this setting. It does not establish reliable global policy ranking or persistent improvements in executed wealth.

10 Conclusion

Common-state replay exposes retrospective trading opportunities, while past-only predictors test which continuation contrasts could have been anticipated. Some registered settings show positive selected-decision evidence, but no continuous selective-repair specification has an entirely positive reported interval for executed wealth. Controlled synthetic tests separate informative decision-time signals from clairvoyance-only opportunity. The 60-day labels remain screening surrogates, and the empirical calibration rule has no established finite-sample safety guarantee for its multi-candidate choice. The evidence supports studying information-recoverable local regret; sustained portfolio gains require separate validation.

A Supporting Theory and Sensitivity Results

This appendix collects standard identities and secondary consequences used to interpret the main results. They are separated from the main theoretical chain because they are not specific mathematical innovations of the repair procedure.

A.1 Oracle-gap decomposition

Let $J(\pi)$ be population expected utility under a fixed evaluation design, let $\pi^0 \in \Pi_{\mathcal{F}}$, and let Π_{O} be a broader comparator class that may use hindsight information while obeying the same action and accounting constraints. Define

$$V_{\mathcal{F}} = \sup_{\pi \in \Pi_{\mathcal{F}}} J(\pi), \qquad V_{\mathrm{O}} = \sup_{\pi \in \Pi_{\mathrm{O}}} J(\pi).$$

Then, whenever the values are finite,

$$V_{\mathrm{O}} - J(\pi^0) = \{V_{\mathrm{O}} - V_{\mathcal{F}}\} + \{V_{\mathcal{F}} - J(\pi^0)\}, \tag{23}$$

with both terms nonnegative. When $V_{\mathrm{O}} > J(\pi^0)$, the ratio

$$\rho_{\mathrm{info}} = \frac{V_{\mathcal{F}} - J(\pi^0)}{V_{\mathrm{O}} - J(\pi^0)} \in [0, 1] \tag{24}$$

measures the fraction of the population oracle gap attainable within the decision-time policy class. It is distinct from the empirical synthetic recovered fraction.

A.2 Information refinement

For square-integrable advantage Δ and information sigma-field $\mathcal{H}$, define

$$\mathcal{R}(\mathcal{H}) = \frac{\mathrm{Var}(\mathbb{E}[\Delta \mid \mathcal{H}])}{\mathrm{Var}(\Delta)}$$

when $\mathrm{Var}(\Delta) > 0$. If Y_{L} is a garbling of Y_{H} so that $\Delta \to Y_{\mathrm{H}} \to Y_{\mathrm{L}}$ is Markov, then the tower property and total-variance identity give

$$\mathrm{Var}(\mathbb{E}[\Delta \mid Y_{\mathrm{L}}]) \le \mathrm{Var}(\mathbb{E}[\Delta \mid Y_{\mathrm{H}}]). \tag{25}$$

Thus more informative observations weakly enlarge the Bayes-recoverable variation, without guaranteeing monotonic performance for a fitted finite-sample predictor or gate.

For the Gaussian signal family $Y_s = sZ + \epsilon_s$ with $\epsilon_s \sim N(0,1)$ independent of (Δ, Z, V) and observation $X_s = (V, Y_s)$, $0 \le s_1 \le s_2$ implies that X_{s_1} is a garbling of X_{s_2}. Indeed, with $a = s_1/s_2$ and independent $\xi \sim N(0,1)$, $\widetilde{Y}_{s_1} = aY_{s_2} + \sqrt{1-a^2}\xi$ has the same conditional law as Y_{s_1}. Hence $\mathcal{R}(X_{s_1}) \le \mathcal{R}(X_{s_2})$.

A.3 Distribution-shift sensitivity for harm–coverage

Condition on the fitted predictor and calibration output so that L is fixed. If

$$P_{\mathrm{cal}}(\Delta < L) \le \alpha, \qquad \mathrm{TV}(P_{\mathrm{test}}, P_{\mathrm{cal}}) \le \eta,$$

then $P_{\mathrm{test}}(\Delta < L) \le \alpha + \eta$. Therefore, for $C_{\mathrm{test}} = P_{\mathrm{test}}(L > \delta) > 0$,

$$P_{\mathrm{test}}(\Delta < \delta \mid L > \delta) \le \min\left\{1, \frac{\alpha + \eta}{C_{\mathrm{test}}}\right\}. \tag{26}$$

The radius η is a sensitivity parameter, not an estimated guarantee in the market experiment.

Table 3: Held-out-asset selective repair. Reported intervals are paired 95% moving-block-bootstrap intervals. Harm and coverage are decision-level proportions among eligible test decisions.

Model	Incumbent	Local advantage [95% CI]	Harm	Coverage	Path $\Delta \log U$ [95% CI]
GBT	BuyHold	-0.0045 [-0.0317, 0.0113]	0.503	0.027	-0.0118 [-0.0300, 0.0012]
GBT	Donchian	0.0755 [0.0563, 0.0925]	0.152	0.102	0.0281 [-0.0155, 0.0739]
GBT	Momentum	0.0281 [0.0194, 0.0406]	0.091	0.162	0.0075 [-0.0222, 0.0393]
GBT	RSI	0.0562 [0.0030, 0.0836]	0.312	0.043	0.0051 [-0.0195, 0.0286]
GBT	SMA	0.0341 [0.0244, 0.0506]	0.165	0.150	0.0087 [-0.0226, 0.0392]
GBT	VolTarget	0.0141 [0.0055, 0.0255]	0.311	0.088	0.0029 [-0.0167, 0.0245]
Ridge	BuyHold	-0.0340 [-0.0626, 0.0000]	0.550	0.006	0.0001 [-0.0031, 0.0034]
Ridge	Donchian	0.0767 [0.0395, 0.0910]	0.156	0.072	0.0096 [-0.0315, 0.0508]
Ridge	Momentum	0.0277 [0.0224, 0.0350]	0.086	0.187	0.0186 [-0.0124, 0.0535]
Ridge	RSI	-0.0469 [-0.1036, -0.0143]	0.802	0.026	-0.0128 [-0.0337, 0.0074]
Ridge	SMA	0.0275 [0.0170, 0.0386]	0.169	0.185	0.0072 [-0.0283, 0.0444]
Ridge	VolTarget	0.0329 [0.0209, 0.0410]	0.130	0.086	0.0174 [0.0009, 0.0345]

A.4 Coverage-limited gain and plug-in ranking

For deployment indicator d with coverage $C = \mathbb{P}(d = 1) > 0$,

$$\mathbb{E}[d\Delta] = C\,\mathbb{E}[\Delta \mid d = 1]. \tag{27}$$

This identity clarifies why a selective gate can show sizeable conditional advantage yet have a small aggregate effect.

For a separate global-ranking observation, suppose conditional policy utilities $m_k(s)$ have estimates satisfying $\max_k |\widehat{m}_k(s) - m_k(s)| \le \varepsilon$. If k^* and $\widehat{k}$ maximize the true and estimated utilities, respectively, then

$$m_{k^*}(s) - m_{\widehat{k}}(s) \le 2\varepsilon. \tag{28}$$

This elementary plug-in bound illustrates why persistent global policy ranking demands stronger uniform accuracy than incumbent-relative repair with abstention.

B Additional Empirical and Synthetic Details

The following material uses the same frozen registry and publication bundle as the main text.

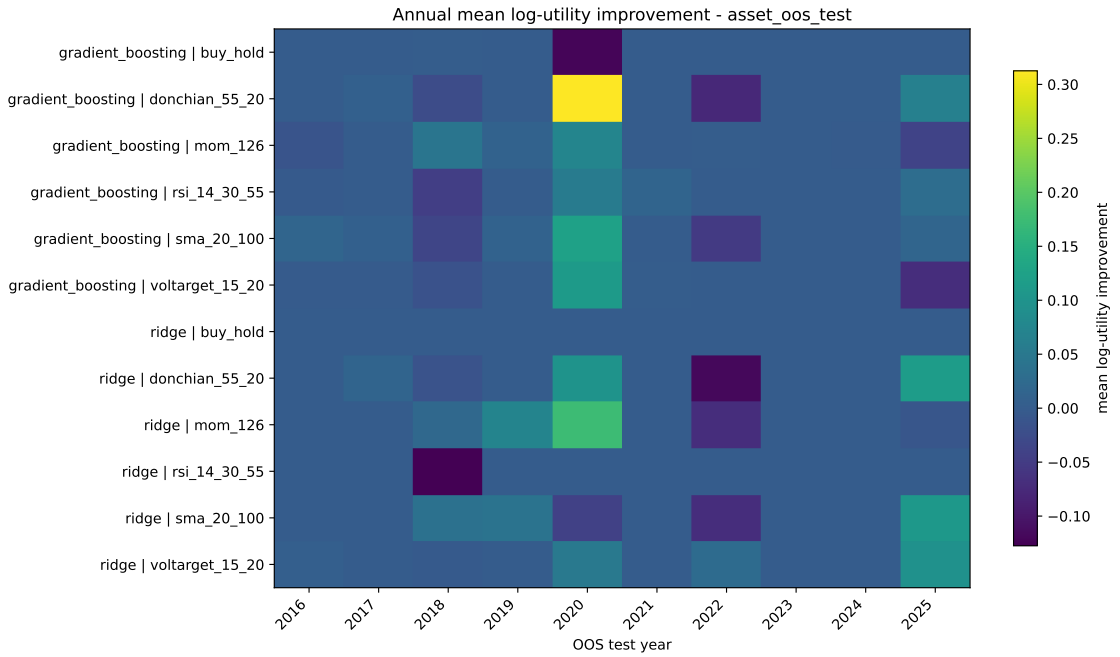
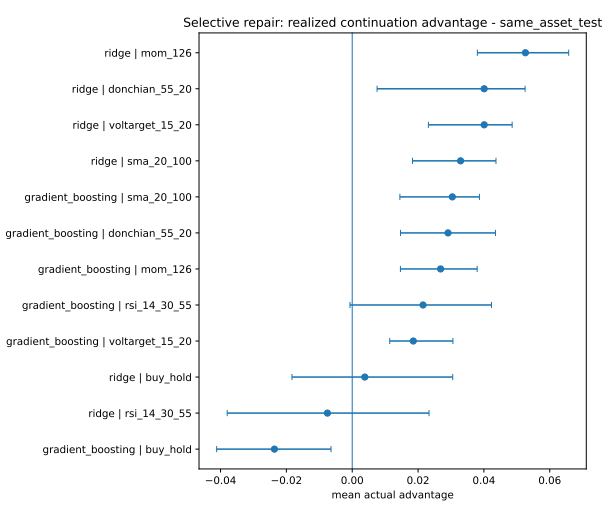


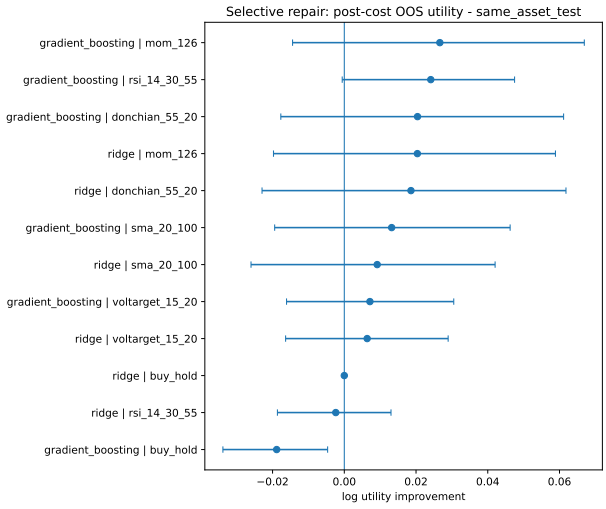
Figure 4: Annual held-out-asset mean log-utility improvement for selective repair. The heatmap emphasizes temporal heterogeneity: aggregate effects are not produced uniformly across test years or policy/model combinations.

B.1 Complete held-out selective-repair table

B.2 Annual and same-asset evidence



(a) Decision-level continuation advantage.



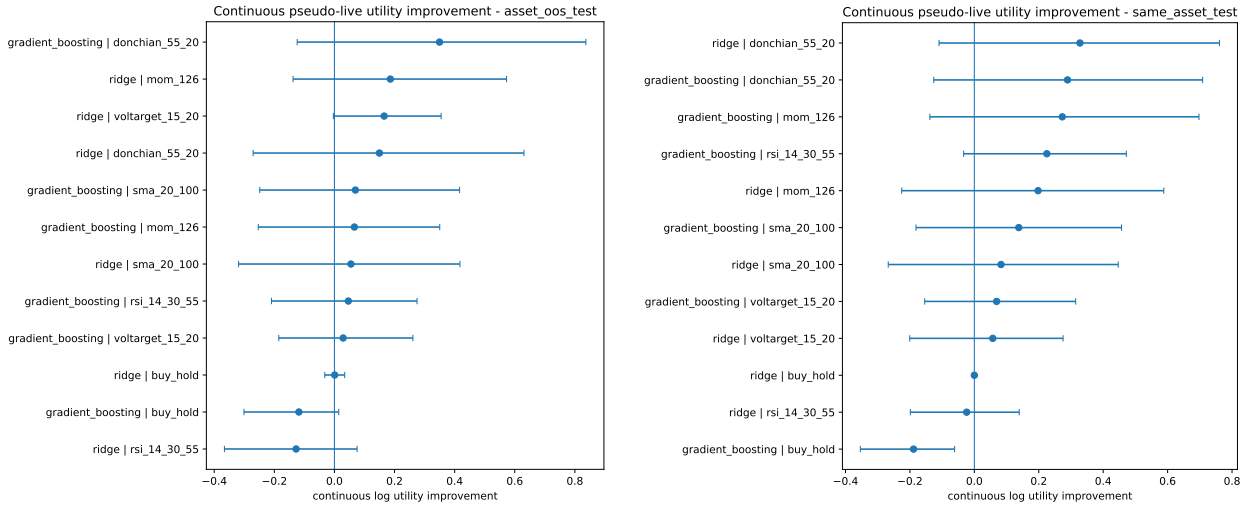
(b) Fold-local path utility.

Figure 5: Same-asset selective-repair evidence. The local/path asymmetry is also present on development assets.

Table 4: Same-asset selective repair. Reported intervals are paired 95% moving-block-bootstrap intervals.

Model	Incumbent	Local advantage [95% CI]	Harm	Coverage	Path $\Delta \log U$ [95% CI]
GBT	BuyHold	-0.0236 [-0.0412, -0.0064]	0.520	0.047	-0.0189 [-0.0338, -0.0046]
GBT	Donchian	0.0291 [0.0147, 0.0435]	0.214	0.101	0.0204 [-0.0177, 0.0611]
GBT	Momentum	0.0268 [0.0146, 0.0380]	0.229	0.195	0.0266 [-0.0144, 0.0669]
GBT	RSI	0.0215 [-0.0007, 0.0423]	0.371	0.052	0.0241 [-0.0006, 0.0475]
GBT	SMA	0.0304 [0.0145, 0.0386]	0.254	0.139	0.0132 [-0.0194, 0.0462]
GBT	VolTarget	0.0186 [0.0114, 0.0306]	0.307	0.114	0.0071 [-0.0161, 0.0305]
Ridge	BuyHold	0.0038 [-0.0183, 0.0305]	0.053	0.017	0.0000 [0.0000, 0.0000]
Ridge	Donchian	0.0401 [0.0075, 0.0525]	0.266	0.082	0.0186 [-0.0229, 0.0618]
Ridge	Momentum	0.0526 [0.0380, 0.0658]	0.208	0.194	0.0204 [-0.0197, 0.0589]
Ridge	RSI	-0.0075 [-0.0380, 0.0233]	0.494	0.036	-0.0024 [-0.0186, 0.0130]
Ridge	SMA	0.0329 [0.0183, 0.0437]	0.204	0.206	0.0092 [-0.0260, 0.0420]
Ridge	VolTarget	0.0401 [0.0231, 0.0486]	0.135	0.084	0.0064 [-0.0164, 0.0290]

B.3 Continuous pseudo-live selective repair



(a) Held-out assets.

(b) Same assets.

Figure 6: Continuous 2016–2025 pseudo-live utility effects. No selective-repair group has a 95% interval entirely above zero.

B.4 Synthetic design and full registered summary

Let $Z_t \in \{\text{trend}, \text{mean-reversion}, \text{noise}, \text{jump}\}$ follow

$$P = \begin{pmatrix} 0.970 & 0.010 & 0.015 & 0.005 \\ 0.010 & 0.970 & 0.015 & 0.005 \\ 0.020 & 0.020 & 0.950 & 0.010 \\ 0.120 & 0.120 & 0.160 & 0.600 \end{pmatrix}. \tag{29}$$

In the recoverable world,

$$r_{t+1} = \mu_{Z_t} + \phi_{Z_t} r_t + \sigma_{Z_t} \varepsilon_{t+1} + J_{t+1}, \tag{30}$$

with $\phi = (0.75, -0.75, 0, 0)$, $\mu = (3 \times 10^{-4}, 3 \times 10^{-4}, 0, 0)$, $\sigma = (0.0035, 0.0035, 0.0050, 0.0060)$, and an additional $N(0, 0.02^2)$ innovation in the jump regime with probability 0.10. Decision-time regime

information is observed through

$$Y_t(s) = s\, g(Z_t) + \eta_t, \qquad g(Z_t) \in \{1, -1, 0, 0\}, \qquad \eta_t \sim N(0, 1), \tag{31}$$

for $s \in \{0, 0.25, 0.5, 1, 2, 4\}$. The feature vector also includes a bounded signed-return state and absolute current return.

In the clairvoyance-only world,

$$r_{t+1} = \epsilon_{t+1} + J_{t+1}, \qquad \epsilon_{t+1} \sim N(0, 0.005^2), \tag{32}$$

where $J_{t+1} \sim N(0, 0.04^2)$ with probability 0.10 and is otherwise zero, independently of regime and decision-time features. Increasing s therefore improves an irrelevant regime proxy without revealing future advantage.

The baseline holds risky exposure when the current return is nonnegative; the candidate holds risky exposure when it is negative. Their 20-day continuation utilities include 2 bps per side. Each experiment contains 3600 business-day observations and is repeated 12 times. A chronological 60%/20%/20% train/calibration/test split is used with a full 20-day purge at both boundaries. Ridge penalty is 1.0, and the lower score uses the calibration residual 0.10 lower-tail quantile.

Table 5: Controlled synthetic identifiability and falsification results. Entries are means across 12 independent replicates. Deployed gain is $\mathbb{E}[d_t \Delta_t]$; oracle gain is $\mathbb{E}[(\Delta_t)_+]$; recovered fraction is their ratio.

World	s	Spearman	Coverage	Deploy gain	Oracle gain	Recovered
Clairvoyance-only	0.00	0.008	0.000	0.0000	0.0245	0.000
Clairvoyance-only	0.25	0.017	0.000	0.0000	0.0245	0.000
Clairvoyance-only	0.50	0.012	0.000	0.0000	0.0245	0.000
Clairvoyance-only	1.00	0.005	0.000	0.0000	0.0245	0.000
Clairvoyance-only	2.00	0.003	0.000	0.0000	0.0245	0.000
Clairvoyance-only	4.00	0.010	0.000	0.0000	0.0245	0.000
Recoverable	0.00	0.067	0.000	0.0000	0.0174	0.000
Recoverable	0.25	0.160	0.000	0.0000	0.0174	0.000
Recoverable	0.50	0.268	0.0002	-3.1×10^{-6}	0.0174	−0.0002
Recoverable	1.00	0.435	0.0068	0.0002	0.0174	0.0129
Recoverable	2.00	0.569	0.0284	0.0011	0.0174	0.0639
Recoverable	4.00	0.609	0.0483	0.0019	0.0174	0.1070

The controlled DGP is separate from the real-market experiment. Synthetic results test identifiability and abstention behavior; they are not evidence of market profitability.

C Reproducibility and Audit Contract

The manuscript is tied to the frozen Phase-12 publication bundle. The experimental registry fixes the random seed (20260924), asset split, six policy definitions, $H = 60$, execution timing, 5 bps per-side transaction cost, starting-exposure grid, feature contract, model classes, walk-forward boundaries, calibration rule, deployment margin, and bootstrap settings. Phase-12 results were not used to retune these quantities.

Feature and model specification. The past-only feature vector contains log momentum at 5, 20, 60, and 126 days; annualized volatility at 10, 20, and 60 days; drawdown from 20-, 60-, and 252-day

rolling maxima; SMA(20)/SMA(100)−1 and SMA(50)/SMA(200)−1; 14-day RSI; and current starting exposure. Missing or non-finite values are replaced by the training median, then centered and scaled with training-only moments. Ridge uses penalty 1.0. Histogram gradient boosting uses learning rate 0.05, 80 iterations, at most 15 leaves, minimum leaf size 50, and L_2 regularization 0.001; early stopping is disabled. Auxiliary predictability diagnostics use the training-only 0.90 regret quantile and logistic regression with $C = 1.0$ and at most 300 iterations; these diagnostics do not enter the primary repair rule.

Chronology and checks. A continuation label is eligible only after its complete future window has been realized. The registered purge is 60 trading days with a further 5-day embargo at fitting boundaries. Preprocessing is fitted under the same membership rule. Asset-OOS tickers never enter training or calibration. Automated checks enforce close-to-next-open timing, shared accounting, corporate-action handling, label availability, train-only preprocessing, held-out-asset isolation, oracle isolation, determinism, and invariance of pseudo-live deployment to removal of test-period hindsight artifacts.

Inference artifacts. Paired circular moving-block bootstrap inference uses block length 20 and 1000 resamples. Manuscript figures and tables are copied from the frozen publication bundle; compact tables used in the main text are formatting views of those frozen outputs rather than recomputed analyses.

Data provenance and code reuse. The supplied experimental code downloads daily market observations with `yfinance` (Yahoo Finance interface), requesting price bars and corporate-action information, with per-symbol request parameters and UTC retrieval timestamps recorded by its download module. Readers should not equate current vendor downloads with immutable point-in-time data: provider revisions, ticker-universe selection, adjustment conventions, and corporate-action reconstruction may affect a new extraction. The frozen raw-data hashes and exact download timestamps should be verified before a fuller data-provenance release; no unverified extraction date or institutional data provenance is asserted here.

A curated public code snapshot is provided at https://github.com/SequentialDecisionResearch/Recoverable-Hindsight-Regret. It contains the Phase 0–12 entry-point scripts and Python modules, experiment and policy YAML files in the runnable `config/` directory, representative tests, and seven archived aggregate CSV result tables. The public repository is a partial reproducibility package: it does not include the original raw market-data snapshots or all intermediate outputs. The separate LaTeX manuscript archive carries the frozen publication bundle and figure/table artifacts. Neither building this PDF nor inspecting the released code constitutes independent numerical reproduction of the reported market results.

D Meta-Selector Stress Test

The global meta-selector is a secondary stress test rather than a primary method. Let c_t be the currently selected policy identity and $\lambda \geq 0$ a fixed policy-switch penalty. It proposes

$$\widetilde{j}_t \in \arg\max_j \left\{ \widehat{U}_t^{(j)} - \lambda \mathbf{1}\{j \neq c_t\} \right\}, \tag{33}$$

and switches from c_t only if $\widetilde{j}_t \neq c_t$ and

$$L_t^{c_t \to \widetilde{j}_t} - \lambda > \delta. \tag{34}$$

Table 6: Meta-selection stress test. The conservative selector is compared with a naive global selector. No fold-local meta-selector interval is positive at 95%; one continuous meta-selector interval is significantly negative.

Split	Strategy	Model	Path $\Delta \log U$ [95% CI]	Coverage	Continuous $\Delta \log U$ [95% CI]
Asset-OOS	Meta	GBT	0.0013 [-0.0177, 0.0223]	0.004	-0.2742 [-0.6680, 0.1122]
Asset-OOS	Meta	Ridge	-0.0002 [-0.0135, 0.0122]	0.000	-0.0138 [-0.1635, 0.1187]
Asset-OOS	Naive meta	GBT	-0.0231 [-0.0586, 0.0117]	0.220	-0.2023 [-0.5946, 0.1487]
Asset-OOS	Naive meta	Ridge	-0.0054 [-0.0166, 0.0037]	0.083	-0.0542 [-0.1700, 0.0423]
Same-asset	Meta	GBT	-0.0159 [-0.0395, 0.0096]	0.005	-0.3016 [-0.5459, -0.0210]
Same-asset	Meta	Ridge	-0.0016 [-0.0067, 0.0015]	0.000	-0.0159 [-0.0669, 0.0155]
Same-asset	Naive meta	GBT	-0.0194 [-0.0487, 0.0111]	0.221	-0.1913 [-0.5081, 0.1071]
Same-asset	Naive meta	Ridge	-0.0028 [-0.0129, 0.0083]	0.090	-0.0276 [-0.1426, 0.0900]

A successful switch updates the policy identity. Transaction costs remain those charged by the execution engine; λ is only a stability penalty. This construction requires persistent ranking across the entire policy family, a stronger condition than incumbent-relative repair.

The conservative gate sharply reduces coverage, especially for Ridge, while the naive selector switches much more frequently. Neither variant provides positive 95% fold-local evidence. The significantly negative continuous same-asset gradient-boosting result is retained in the main evidence set.

Data and Code Availability

A public research-code snapshot is available at `https://github.com/SequentialDecisionResearch/Recoverable-Hindsight-Regret`. It includes multi-phase Python sources, selected tests, experiment and policy configurations, documentation, and seven archived aggregate result tables. The separate manuscript source archive contains the frozen figures and publication outputs used to compile this paper. Market observations were requested through the `yfinance` interface (Yahoo Finance access); raw third-party historical prices are not redistributed in the GitHub repository. Current vendor downloads may differ from the frozen author-side data and do not constitute a certified point-in-time database. Some intermediate artifacts and original data snapshots are absent from the public repository, and an independent end-to-end numerical reproduction of the reported market results has not been established.

References

David H. Bailey and Marcos López de Prado. The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *The Journal of Portfolio Management*, 40(5):94–107, 2014.

David H. Bailey, Jonathan Borwein, Marcos López de Prado, and Qiji Jim Zhu. The probability of backtest overfitting. *The Journal of Computational Finance*, 20(4), 2017.

Omar Besbes, Yuri Fonseca, and Ilan Lobel. Online learning from optimal actions. In *Proceedings of the 34th Conference on Learning Theory*, PMLR 134:586–586, 2021.

Thomas M. Cover. Universal portfolios. *Mathematical Finance*, 1(1):1–29, 1991.

Miroslav Dudík, John Langford, and Lihong Li. Doubly robust policy evaluation and learning. In *Proceedings of the 28th International Conference on Machine Learning*, pages 1097–1104, 2011.

Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems 30*, pages 4878–4887, 2017.

Mohammad Ghavamzadeh, Marek Petrik, and Yinlam Chow. Safe policy improvement by minimizing robust baseline regret. In *Advances in Neural Information Processing Systems 29*, pages 2298–2306, 2016.

Isaac Gibbs and Emmanuel J. Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems 34*, 2021.

Peter R. Hansen. A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4):365–380, 2005.

Hans R. Künsch. The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17(3):1217–1241, 1989.

Romain Laroche, Paul Trichelair, and Rémi Tachet des Combes. Safe policy improvement with baseline bootstrapping. In *Proceedings of the 36th International Conference on Machine Learning*, PMLR 97:3652–3661, 2019.

Dimitris N. Politis and Joseph P. Romano. The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994.

Yaniv Romano, Evan Patterson, and Emmanuel J. Candès. Conformalized quantile regression. In *Advances in Neural Information Processing Systems 32*, 2019.

Abhishek Sharma, Leo Benac, Sonali Parbhoo, and Finale Doshi-Velez. Decision-point guided safe policy improvement. In *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics*, PMLR 258:2935–2943, 2025.

Philip Thomas, Georgios Theocharous, and Mohammad Ghavamzadeh. High confidence policy improvement. In *Proceedings of the 32nd International Conference on Machine Learning*, PMLR 37:2380–2388, 2015.

Halbert White. A reality check for data snooping. *Econometrica*, 68(5):1097–1126, 2000.